

FIITJEE

ALL INDIA TEST SERIES

FULL TEST – V

JEE (Main)-2024

TEST DATE: 17-01-2024

Time Allotted: 3 Hours

Maximum Marks: 300

General Instructions:

- The test consists of total 90 questions.
- Each subject (PCM) has 30 questions.
- This question paper contains **Three Parts**.
- **Part-A** is Physics, **Part-B** is Chemistry and **Part-C** is Mathematics.
- Each part has only two sections: **Section-A** and **Section-B**.
- **Section – A** : Attempt all questions.
- **Section – B** : Do any five questions out of 10 questions.

Section-A (01 – 20, 31 – 50, 61 – 80) contains 60 multiple choice questions which have **only one correct answer**. Each question carries **+4 marks** for correct answer and **–1 mark** for wrong answer.

Section-B (21 – 30, 51 – 60, 81 – 90) contains 30 Numerical based questions. The answer to each question is rounded off to the nearest integer value. Each question carries **+4 marks** for correct answer and **–1 mark** for wrong answer.

Physics

PART – A

SECTION – A (One Options Correct Type)

This section contains **20 multiple choice questions**. Each question has **four choices** (A), (B), (C) and (D), out of which **ONLY ONE** option is correct.

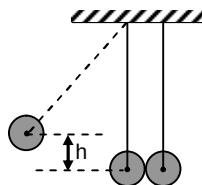
- If a wheel is rolling without slipping on a horizontal surface at a constant linear speed then the point of contact has

(A) backward acceleration (B) zero acceleration
(C) an upward acceleration (D) a downward acceleration
- The rms value of the electric field of the light coming from the sun is 720 N/C. The average total energy density of the electromagnetic wave is

(A) $4.58 \times 10^{-6} \text{ J/m}^3$ (B) $6.37 \times 10^{-9} \text{ J/m}^3$
(C) $81.35 \times 10^{-12} \text{ J/m}^3$ (D) $3.3 \times 10^{-3} \text{ J/m}^3$
- A person sitting firmly over a rotating stool has his arms folded with two identical balls. If he stretched his arms along with balls and then the work done by him

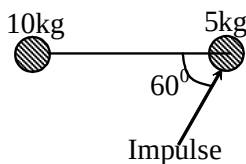
(A) zero (B) positive
(C) negative (D) any of these
- In the figure, pendulum bob on left side is pulled aside to a height h from its initial position. After it is released it collides with the right pendulum bob at rest, which is of the same mass. After the collision the two bobs stick together and raise to a height:

(A) $\frac{3h}{4}$ (B) $\frac{2h}{3}$
(C) $\frac{h}{2}$ (D) $\frac{h}{4}$


- Three particles each of mass m are placed at the corners of an equilateral triangle of side d . The potential energy of the systems is

(A) $\frac{3Gm^2}{d}$ (B) $\frac{Gm^2}{d}$
(C) $\frac{-3Gm^2}{d}$ (D) none of these.
- Two point masses connected by an ideal string are placed on a smooth horizontal surface as shown in the diagram. A sharp impulse of 10 kg-m/s is given to the 5 kg mass. The velocity of the 10 kg mass will be

(A) $\frac{2}{3} \text{ m/s}$ (B) $\frac{1}{3} \text{ m/s}$
(C) 2 m/s (D) zero

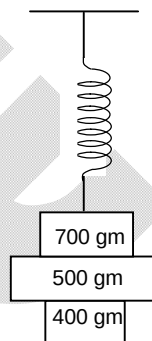


7. A satellite of mass m revolves around the earth of radius R at a height x from its surface. If g is acceleration due to gravity at surface then orbital speed is

(A) gx (B) $\frac{gR}{R-x}$
 (C) $\frac{gR^2}{R+x}$ (D) $\left(\frac{gR^2}{R+x}\right)^{\frac{1}{2}}$

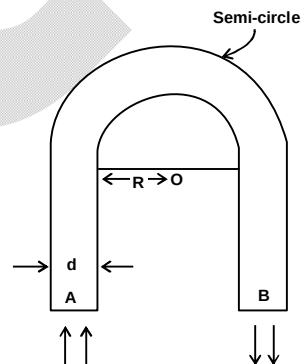
8. Three masses 700 gm, 500 gm and 400 gm are suspended at the end of a massless spring as shown, when 700 gm is removed system oscillates with a period 3 sec. What will be its period when 500 gm is also removed

(A) 1 sec (B) 2 sec
 (C) 3 sec (D) $\sqrt{\frac{12}{5}}$ sec

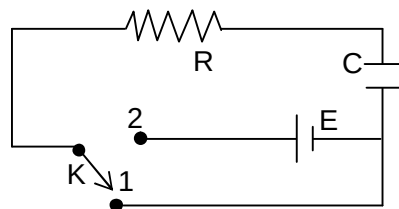


9. A glass rod ($\mu = 1.5$) of square cross-section is bent into the shape shown in the figure. A parallel beam of light falls perpendicularly on the plane flat surface A. Referring to the diagram, d is the width of a side and R is the radius of inner semi-circle. The maximum value of ratio (d/R) so that all light entering the glass through surface A emerge from the glass through surface B is

(A) $1/2$ (B) $2/3$
 (C) $3/2$ (D) $2/1$



10. In the shown circuit involving a resistor of resistance $R \Omega$, capacitor of capacitance C farad and an ideal cell of emf E volt, the capacitor is initially uncharged and the key is in position 1. At $t = 0$ second the key is pushed to position 2 for $t_0 = RC$ seconds and then key is pushed back to position 1 for $t_0 = RC$ seconds. This process is repeated again and again. Assume the time taken to push key from position 1 to 2 and vice versa to be negligible.



The charge on capacitor at $t = 2RC$ second is

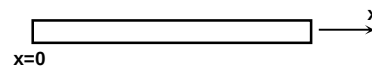
(A) CE (B) $CE\left(1 - \frac{1}{e}\right)$
 (C) $CE\left(\frac{1}{e} - \frac{1}{e^2}\right)$ (D) $CE\left(1 - \frac{1}{e} + \frac{1}{e^2}\right)$

11. A particle starts its motion from rest and moves with constant acceleration for time t_1 and then it retards with constant rate for time t_2 until it comes to rest. Then the ratio of maximum speed and average speed during the complete motion will be

(A) $2 : 1$ (B) $1 : 2$
 (C) $t_1 : t_2$ (D) $t_2 : t_1$

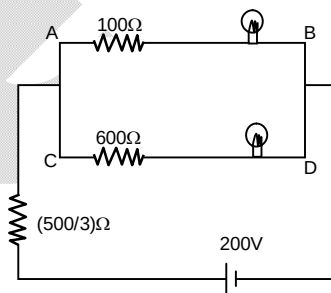
12. An open pipe of length 47 cm has fundamental frequency 340 Hz. What is the diameter of the tube if velocity of sound in air is 340 m/sec?
- (A) 2 cm (B) 5 cm
(C) 8 cm (D) 10 cm

13. If along a uniform rod of length ℓ carrying current I , the voltage V changes with position x along the length of the rod such that $dV/dx = -k$, where k is a positive number, then the resistance of the rod is



- (A) $k\ell/2I$ (B) $k\ell/I$
(C) $I/k\ell$ (D) $k\ell$
14. Two balls of equal mass have a head on collision with speed 6 m/s. If the coefficient of restitution is $1/3$, the velocity of each ball after impact will be
- (A) 18 m/s (B) 2 m/s
(C) 6 m/s (D) 4 m/s

15. Two bulbs 25W, 100V (upper bulb in figure) and 100W, 200V (lower bulb in figure) are connected in the circuit as shown in figure. Choose the correct answer.

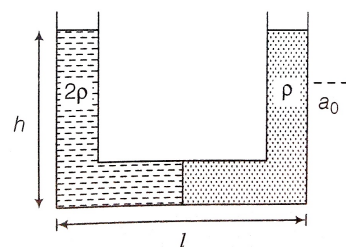


- (A) Heat lost per second in the circuit will be 80J
(B) Ratio of heat produced per second in bulbs will be 1: 1
(C) Ratio of heat produced in branch AB to branch CD will be 1:2
(D) Current drawn from the cell is 0.2 A

16. The first overtone of an open organ pipe beats with the first overtone of a closed organ pipe with a beat frequency of 2.2 Hz. The fundamental frequency of the closed organ pipe is 110 Hz. The lengths of the pipes are

- (A) 1.1 m, 0.9 m (B) 1.2 m, 0.8 m
(C) 1.0067 m, 0.9937 m (D) 2.0067 m, 0.9937 m
17. A particle is projected horizontally from the top of a tower with a velocity v_0 . If v be its velocity at any instant, then the radius of curvature of the path of the particle at the point (where the particle is at that instant) is directly proportional to
- (A) v^3 (B) v^2
(C) v (D) $\frac{1}{v}$

18. A U-tube of base length l filled with same volume of two liquids of densities p and $2p$ is moving with an acceleration a_0 on the horizontal plane. If the height difference between the two surfaces (open to atmosphere) becomes zero. What will be the height h ?



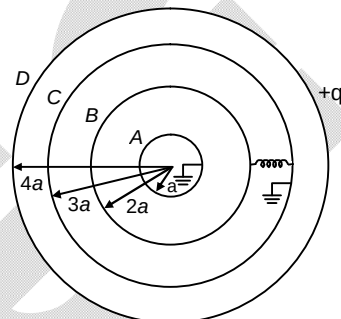
- (A) $h = \frac{3a_0 l}{2g}$ (B) $h = \frac{a_0 l}{2g}$
(C) $h = \frac{3a_0 l}{g}$ (D) $h = \frac{2a_0 l}{3g}$

19. A tunnel is dug along diameter inside the surface of earth and a particle is projected from the centre of tunnel. The minimum velocity of particle such that it escape out from the earth's gravitation field is (Radius of earth = R_e)

- (A) $\sqrt{2gR_e}$ (B) $\sqrt{\frac{3}{2}gR_e}$
 (C) $\sqrt{3gR_e}$ (D) $\sqrt{\frac{5}{2}gR_e}$

20. Four concentric conducting shells A, B, C and D are arranged as shown in figure. A charge $+q$ is given to the outer most shell D. Shell B and C are connected by conducting wire while Shell A and C is earthed. The equivalent capacitance of system is

- (A) $64\pi\epsilon_0 a$
 (B) $16\pi\epsilon_0 a$
 (C) $4\pi\epsilon_0 a$
 (D) none of these



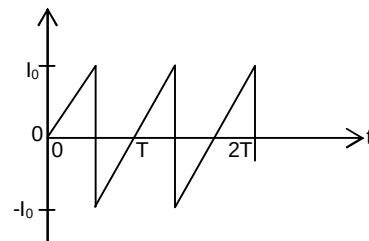
SECTION – B

(Numerical Answer Type)

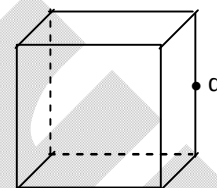
This section contains **10** Numerical based questions. The answer to each question is rounded off to the nearest integer value.

21. Two different coils have self inductances $L_1 = 8$ mH and $L_2 = 2$ mH. The current in one coil is increased at a constant rate. The current in the second coil is also increased at the same constant rate. At this time the induced voltage in the first coil is V_1 and for the second coil at the same instant V_2 respectively. Then V_1/V_2 is
22. A non-viscous liquid of constant density 10^3 kg/m³ flows in streamline motion along a vertical tube PQ of variable cross-section. Height of P and Q are 2 m and 2.5 m respectively. Area of tube at Q = 3 time area of tube at P. Find the work done per unit volume by pressure in kJ/m³ as liquid flows from P to Q. Speed of liquid at P is 3 m/s ($g = 10$ m/s²).
23. In a moving coil galvanometer, torque on the coil can be expressed as $\tau \propto i$, where i is current through the wire. The rectangular coil of the galvanometer having number of turns N , area A and moment of inertia I is placed in magnetic field B . The torsion constant of spring in the galvanometer is C and charge Q is passed through the coil almost instantaneously. If the maximum angle through which coil deflected is $\frac{KNABQ}{\sqrt{4CI}}$. (Ignore the damping in mechanical oscillations), then the value of K is
24. A charge Q is divided into two parts of magnitude q and $Q - q$. If the coulomb repulsion between them when they are separated at some distance is to be maximum, the ratio of $\frac{Q}{q}$ should be

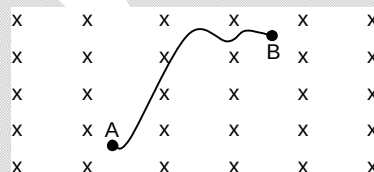
25. For a circuit, current variation with time is shown in figure, corresponding rms current is $\frac{I_0}{\sqrt{n}}$ then find the value of 'n'.



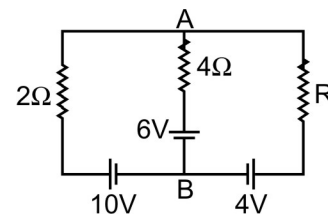
26. The total flux of electric lines of forces of a charge q placed at the mid point of the edge of a side of rectangular box, as shown in figure through the box is $\frac{q}{n\epsilon_0}$, then the value of 'n' is



27. The wire shown in the figure between points A and B carries a current of 10A. A uniform magnetic field of 10T exists in the region which is pointing into the plane in which, wire lies. The coordinates of A and B are (2, 0) m and (0, 2) m respectively and the length of the wire is 10m. If the magnitude of the force acting on the wire is $m \times 100\sqrt{2}$ N. Find value of m.



28. Potential energy of a particle moving along x-axis is given by $U = \frac{x^3}{3} - \frac{9x^2}{2} + 20x$. Find out position of stable equilibrium state.
29. In the given circuit, find the value of R (in Ω) for which current through 4Ω resistor is zero :



30. 2 kg of ice at -22.5°C is mixed with 2.5 kg of water at 25°C in an insulating container. If the specific heat of ice and water are $0.5 \text{ cal/gm}^\circ\text{C}$ and $1 \text{ cal/gm}^\circ\text{C}$ respectively. Latent heat of fusion of ice = 80 cal/gm . Find the amount of water present in the container (In kg).

Chemistry**PART – B****SECTION – A**
(One Options Correct Type)

This section contains **20 multiple choice questions**. Each question has **four choices** (A), (B), (C) and (D), out of which **ONLY ONE** option is correct.

31. The radius of the first orbit of ground state hydrogen atoms is $x \text{ Å}$. What will be the radius of the first orbit of Li^{2+} ion in Å ?

(A) $\frac{x}{6}$
(C) $\frac{x}{9}$

(B) $\frac{x}{3}$
(D) $\frac{x}{4}$

32. $\text{X(g)} + 2\text{Y(g)} \longrightarrow \text{Products}$

The rate equation of above reaction is given as follows:

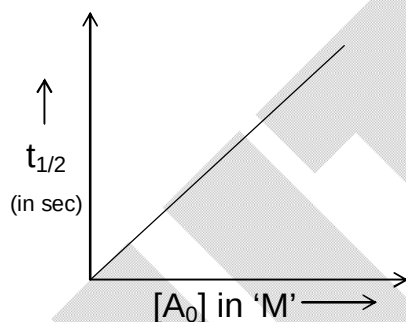
$$\text{Rate} = k[\text{X}][\text{Y}]^{0.5}$$

What is the unit of rate constant of above reaction?

(A) $\text{mol}^{1/2} \text{L}^{-1/2} \text{s}^{-1}$
(C) $\text{mol}^{3/2} \text{L}^{-3/2} \text{s}^{-1}$

(B) $\text{mol}^{-1/2} \text{L}^{1/2} \text{s}^{-1}$
(D) $\text{mol}^{-3/2} \text{L}^{3/2} \text{s}^{-1}$

- 33.

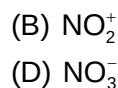
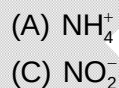


What is the unit of rate constant of the reaction, for which the above graph is given?

(A) s^{-1}
(C) $\text{mol L}^{-1} \text{s}$

(B) $\text{mol L}^{-1} \text{s}^{-1}$
(D) $\text{mol}^{-1} \text{L}^2 \text{s}^{-1}$

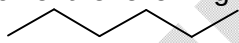
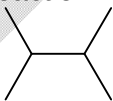
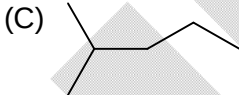
34. Which of the following species has the largest bond angle?



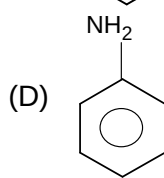
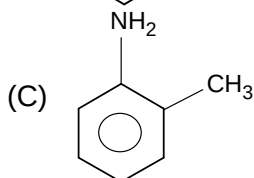
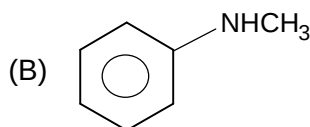
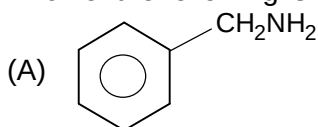
35. What is the oxidation number of iodine in KI_3 ?

(A) -1
(C) $-\frac{1}{3}$

(B) $+1$
(D) $+\frac{1}{3}$

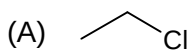
36. $\text{HCN}(\text{aq}) \rightleftharpoons \text{H}^+(\text{aq}) + \text{CN}^-(\text{aq})$
 The ionization constant (K_a) of above acid increases by
 (A) adding more water (B) adding more HCN
 (C) removing H^+ ions by adding bases (D) none of these
37. 100 mL of a sample of hard water requires 25.1 mL of 0.02 N H_2SO_4 for complete reaction. The hardness of water (density 1 g/mL) is:
 (A) 200 ppm (B) 250 ppm
 (C) 251 ppm (D) 258 ppm
38. $\text{P}_4\text{O}_{10} + \text{HNO}_3 \longrightarrow \text{H}_3\text{PO}_4 + (\text{X})$
 In above reaction (X) is
 (A) HNO_2 (B) N_2O_3
 (C) N_2O_5 (D) N_2O_4
39. Which of the following is used as a bleaching agent?
 (A) I_2O_5 (B) HClO
 (C) H_2S (D) IF_5
40. Number of $\text{C}=\text{C}$ bonds present in anthracene is
 (A) 5 (B) 3
 (C) 9 (D) 7
41. $\text{CH}_3\text{CH}_2\text{CH}_2\text{Cl} + \text{CH}_3\underset{\text{Cl}}{\text{CH}}\text{CH}_3 \xrightarrow[\text{Dry ether}]{\text{Na}} \text{Products}$
 Which of the following is not a product of above reaction?
 (A)  (B) 
 (C)  (D) 
42. Which does not behave as a π -acid ligand due to lack of ability to gain electrons from metal ions?
 (A) NO^+ (B) CO
 (C) NH_3 (D) CN^-
43. Which of the following compound has the highest boiling point?
 (A) $\text{CH}_3\text{CH}_2\text{COOCH}_3$ (B) $\text{CH}_3\text{CH}_2\text{CH}_2\text{CONH}_2$
 (C) $\text{CH}_3\text{COOCOCH}_3$ (D) $\text{CH}_3\text{CH}_2\text{COCl}$

44. Which of the following is the strongest base?

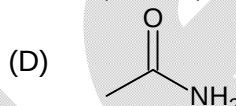
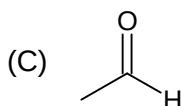


45. $[X] \xrightarrow[\text{(ii) } H^+ (aq)]{\text{(i) } NH_4Cl; KCN (equimolar)} CH_3CH(NH_2)COOH$

Compound [X] is

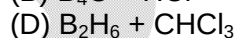
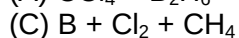
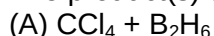


(B)

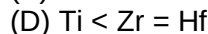
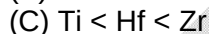
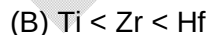
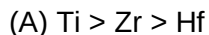


46. $BCl_3 + H_2 + C \xrightarrow{1700-1800^\circ C} \text{Product(s)}$

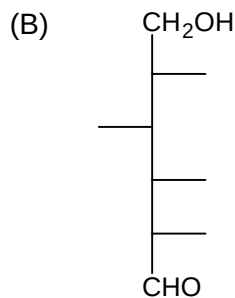
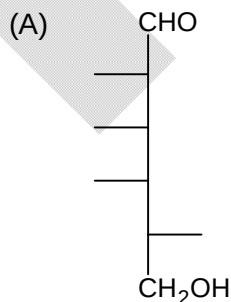
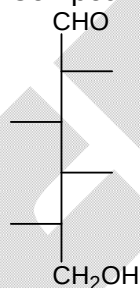
The product(s) of above reaction is/are

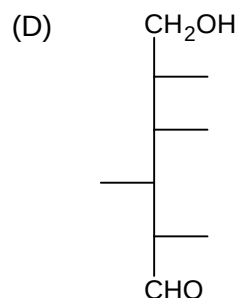
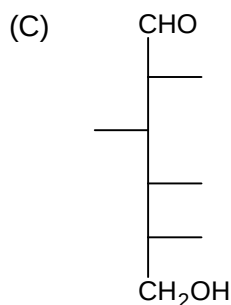


47. Atomic radii of Ti, Zr and Hf vary



48. Compound which gives same osazone as given by





49. Which forms such an alcohol with $\text{CH}_3\text{MgBr}/\text{H}_3\text{O}^+$, which does not undergo oxidation?
 (A) $\text{CH}_3\text{CH}_2\text{CHO}$ (B) HCHO
 (C) $(\text{C}_2\text{H}_5)_2\text{CO}$ (D) $\text{C}_6\text{H}_5\text{CHO}$
50. The functional isomer of which amine is more basic than itself?
 (A) $(\text{CH}_3)_3\text{N}$ (B) CH_3NH_2
 (C) $\text{CH}_3\text{CH}_2\text{NH}_2$ (D) CH_3NHCH_3

SECTION – B

(Numerical Answer Type)

This section contains **10** Numerical based questions. The answer to each question is rounded off to the nearest integer value.

51. How many geometrical isomers are possible for the following complex?
 $[\text{Pt}(\text{NH}_3)(\text{NH}_2\text{OH})(\text{NO}_2)(\text{py})]\text{NO}_2$
52. $\text{Zn(s)} | \text{Zn}^{2+}(0.01 \text{ M}) || \text{H}^+ | \text{H}_2(1 \text{ atm}), \text{Pt}$
 The e.m.f of the above electrochemical cell is 0.6418 V. What is the pH of the cathode half-cell if $E^\circ_{\text{Zn}^{2+}/\text{Zn}} = -0.76 \text{ V}$
53. For a dilute solution containing 2.5 g of a non-volatile non-electrolyte solute in 100 g of water, the elevation in boiling point at 1 atm pressure is 2°C . Assuming concentration of solute is much lower than the concentration of solvent, the vapour pressure (mm of Hg) of solution is (take $K_b = 0.76 \text{ K kg mol}^{-1}$)
54. Metaboric acid is HBO_2 . It is obtained by dehydration of orthoboric acid H_3BO_3 . What is the molar mass of the planar cyclic form of metaboric acid monomer in g mol^{-1} unit?
 [At. mass of H = 1, O = 16, B = 11]
55. Dissociation constant for a weak base BOH is 10^{-5} . What is the value of pH when it is half neutralized by HCl ?
56. What is the minimum sum of (n, ℓ and m) of the unpaired electron of chlorine in ground state? (n, ℓ and m are the quantum numbers)
57. The enthalpy change of a chemical reaction $\text{A(g)} + \text{B(g)} \rightleftharpoons 2\text{C(g)}$ is 70.4 kJ mol^{-1} . At what temperature in Kelvin will the reaction attain equilibrium if the entropies of A, B and C gases respectively are 250, 350 and $400 \text{ J K}^{-1} \text{ mol}^{-1}$?

58. Beryllium pyrosilicate is written as $\text{Be}_x\text{Si}_y\text{O}_z$. What is its molar mass in g mol^{-1} unit?
[At. mass Be = 9, Si = 28, O = 16]
59. If the initial concentration of reactant in a first order reaction is 2M, the half-life is 20 sec, what will be the half-life if the reaction starts with 4M concentration of the reactant?
60. What volume of 1 M HCl should be added to 110 mL of 1 M NaCN, so that the resulting solution will behave as buffer with highest pH?

Mathematics

PART – C

SECTION – A (One Options Correct Type)

This section contains **20 multiple choice questions**. Each question has **four choices** (A), (B), (C) and (D), out of which **ONLY ONE** option is correct.

61. Maximum distance of any point on the curve $2013x^2 - 4008xy + 2013y^2 = 1$ from origin is
 (A) 2013 (B) 9
 (C) 3 (D) $\frac{1}{3}$
62. The relation R on the set $A = \{1, 2, 3, 4, 5, 6\}$ defined by $R = \{(x, y) \mid x, y \in A \text{ and } y = 2x + 1\}$ is
 (A) Reflexive (B) Symmetric
 (C) Transitive (D) Equivalence
63. The expression $P(x) = (\sqrt{x^5 - 1} + x)^7 - (\sqrt{x^5 - 1} - x)^7$ is polynomial of degree
 (A) 16 (B) 18
 (C) 20 (D) 27
64. The number of ways in which three distinct numbers are in A.P. can be selected from the set $\{1, 2, 3, \dots, 24\}$, is equal to:
 (A) 66 (B) 132
 (C) 198 (D) none of these
65. If $[x]$ denotes the integral part of x and $f(x) = [n + p \sin x], 0 < x < \pi, n \in \mathbb{I}$ and p is a prime number, then the number of points, where $f(x)$ is not differentiable is
 (A) $p - 1$ (B) p
 (C) $2p - 1$ (D) $2p + 1$
66. For the hyperbola $xy + 4 = 4x + 2y$ which of the following is not true ?
 (A) $x = 2$ is an asymptote
 (B) $y = 4$ is an asymptote
 (C) $xy + 16 = 4x + 2y$ is the equation of the conjugate hyperbola
 (D) $xy + 12 = 4x + 2y$ is the equation of the conjugate hyperbola
67. The equation of the circle passing through the point of intersection of the curves $(2x+3y+19)(9x+6y-17) = 0$ and $xy = 0$ is
 (A) $x^2 + y^2 + 137x + 63y - 303 = 0$ (B) $4x^2 + 4y^2 + 137x + 63y - 323 = 0$
 (C) $18x^2 + 18y^2 + 137x + 63y - 323 = 0$ (D) None of these

68. If all the real solutions of the equation $4^x - (a-3)2^x + (a-4) = 0$ are non positive, then
(A) $4 < a \leq 5$ (B) $0 < a < 4$
(C) $a > 4$ (D) $a < 3$
69. The number of ordered triplets (a, b, c) , $a, b, c \in \mathbb{N}$, such that $a + b + c \leq 20$ is
(A) less than 100 (B) less than 1000
(C) equal to 1000 (D) more than 1000
70. In a hockey tournament, a total of 153 matches were played. If each team played one match with every other team, the total number of teams that participated in the tournament were
(A) 20 (B) 18
(C) 16 (D) 14
71. Sum of series of $\cot^{-1} \frac{5}{\sqrt{3}} + \cot^{-1} \frac{9}{\sqrt{3}} + \cot^{-1} \frac{15}{\sqrt{3}} + \cot^{-1} \frac{23}{\sqrt{3}} + \dots \infty$ is equal to
(A) $\frac{\pi}{4}$ (B) $\frac{\pi}{3}$
(C) $\frac{\pi}{6}$ (D) $\frac{\pi}{12}$
72. Let $f : [0,1] \rightarrow \mathbb{R}$ be a continuous function then which of the following is maximum value of $\int_0^1 x^2 f(x) dx - \int_0^1 x (f(x))^2 dx$ for all such functions
(A) $\frac{1}{12}$ (B) $\frac{1}{16}$
(C) $\frac{1}{8}$ (D) $\frac{1}{20}$
73. If $f(x)$ is a polynomial function with positive integral coefficients (degree of $f(x) \geq 1$) and for $x \geq 1$, $\frac{f(f[x]+1)}{f([x])}$ is an integer then (where $[.]$ denotes the greatest integer function)
(A) $x \in [1, 2)$ (B) $x \in [2, 3)$
(C) $x \in [3, 4)$ (D) none of these
74. A five digits number has to be formed by using the digits 1, 2, 3, 4 and 5 without repetition such that the even digits occupy odd places. Find the sum of all such possible numbers
(A) 1199988 (B) 1166688
(C) 1266688 (D) None of these

75. The function $f : \mathbb{R} \rightarrow \mathbb{R}$ defined as $f(x) = \ln\left(\sqrt{\sqrt{x^2+1}+x} + \sqrt{\sqrt{x^2+1}-x}\right)$ is:
 (A) one – one and onto both (B) one – one but not onto
 (C) onto but not one – one (D) neither one – one nor onto
76. $\int \frac{\cos x + \sqrt{3}}{1 + 4 \sin\left(x + \frac{\pi}{3}\right) + 4 \sin^2\left(x + \frac{\pi}{3}\right)} dx$ is
 (A) $\frac{\cos x}{1 + 2 \sin\left(x + \frac{\pi}{3}\right)} + c$ (B) $\frac{\sec x}{1 + 2 \sin\left(x + \frac{\pi}{3}\right)} + c$
 (C) $\frac{\sin x}{1 + 2 \sin\left(x + \frac{\pi}{3}\right)} + c$ (D) $\frac{1}{2} \tan^{-1}\left(1 + 2 \sin\left(x + \frac{\pi}{3}\right)\right) + c$
77. If A is skew symmetric matrix and n is odd positive integer then A^n is
 (A) a symmetric matrix (B) a skew symmetric matrix
 (C) a diagonal matrix (D) none of these
78. A plane containing the line $\frac{x-3}{2} = \frac{y-1}{4} = \frac{z-2}{5}$ and its passing through the point (4, 3, 7). The equation of the plane is
 (A) $4x + 8y + 8z = 4$ (B) $4x - 8y - 8z = 4$
 (C) $4x - 2y - 10 = 0$ (D) $4x - 8y + 8z = 4$
79. Let $f : \mathbb{R} \rightarrow \mathbb{R}$, $f(x) = 8x^3 + 3x$. Then $\lim_{x \rightarrow \infty} \frac{f^{-1}(8x) - f^{-1}(x)}{x^{1/3}}$ equals
 (A) 1 (B) $1 - \frac{1}{3^{1/3}}$
 (C) $1 - \frac{1}{2^{1/3}}$ (D) $\frac{1}{2}$
80. Let z_1, z_2, z_3, z_4 be distinct complex numbers when $|z_1| = |z_2| = |z_3| = |z_4| = 1$ Such that $0 \leq \arg(z_1) < \arg(z_2) < \arg(z_3) < \arg(z_4)$ and $\arg(z_2) + \arg(z_4) - \arg(z_1) - \arg(z_3) = \pi$, then $|z_1 z_3 - (z_1 - z_3)(z_2 - z_4)|^2 - |(z_1 - z_3)(z_2 - z_4)|^2$ equals _____
 (A) 1 (B) 2
 (C) 3 (D) 4

SECTION – B

(Numerical Answer Type)

This section contains **10** Numerical based questions. The answer to each question is rounded off to the nearest integer value.

81. The value of the expression $\log_{\sqrt{2}} \log_{1/4} \left(\cos \frac{\pi}{17} \cdot \cos \frac{2\pi}{17} \cdot \cos \frac{4\pi}{17} \cdot \cos \frac{8\pi}{17} \right)^4$ is
82. If normals drawn to $y^2 = 4x$ at $(1, 2)$, $(9, -6)$ and (α, β) are concurrent then $\alpha\beta =$
83. If $z \in \mathbb{C}$ the eccentricity of $|z - (1+2i)| + |z - (-1+2i)| = 4$ is e then $2e$ is
84. The length of latus rectum of the parabola $25[(x-1)^2 + (y-4)^2] = (3x+4y-4)^2$ is
85. The area enclosed by the curve $\max\{|x-1|, |y|\} = k$ is 100, then k is equal to
86. Let $f(x) = (\sin^{-1} x)^3 + (\cos^{-1} x)^3$ and $\max f(x) + \min f(x) = \frac{p}{q} \pi^3$, and $q-p =$
87. If $f''(x) + f'(x) + f^2(x) = x^2$ be differential equation of a curve and let P be the point of maxima then the number of tangents which can be drawn from P to $x^2 - y^2 = a^2$ is/are
88. The expression $\tan 55^\circ \cdot \tan 65^\circ \cdot \tan 75^\circ$ simplifies to $\cot x^\circ$ where $x \in (0, 90)$, then x equals to :
89. The value of $a \in \mathbb{R}$, such that $f: \mathbb{R} \rightarrow \mathbb{R}$ defined by $f(x) = x^3 + x^2 + a \cos x$ is injective
90. Let $f(x)$ be a differentiable function satisfying $f'(x) = f(x)$ with $f(0) = 1$ and $g(x)$ satisfies $f(x) + g(x) = e^x (x+1)^2$. If $\int_0^1 f(x)g(x)dx = ae^2 + b$, then the value of $(a+b)$ is